

Implementation of Project Requirements

Item No.	Feature	Implementation
1	Allow user to enter login details	'login.html' page created containing login form.
	Validate login features	'auth1.js' Node module created to deal with login validation. Users array created to store user credentials with a function to add and authenticate users.
	Website credentials to be 'user@123.com' and 'pass'	'createUser' function from 'auth1.js' called in 'app1.js' to create new user with 'user@123.com' and 'pass'. Post method used on line 44 of 'app1.js' to read user credentials from 'login.html' and render 'checkout.ejs' page if login is successful, or 'loginFailed.ejs' if unsuccessful.
2	Create a form and perform form validation to ensure that text fields not empty and a valid email is entered	'Newsletter sign up' link added to 'Home.html' page, which directs user to sign-up form. First & last name, email address and T&Cs box added to sign up form. Each field marked as required and correct input type set to ensure form inputs are valid. Alert displayed upon pressing submit button if inputs are validated to tell the user that they're signed up.
3	Include a slideshow or carousel which displays a different image each time the page is loaded	Bootstrap carousel added to 'Home.html' on line 41, which displays different images related to the electronics store, such as sale info and pricing endorsements.
4	Allow the user to 'purchase' items from the site	'addToCart' JS function added to 'prodData1.ejs', which saves the product name, QTY and price to browser local storage. JS added to 'checkout.ejs' which reads the information in localStorage, calculates the total price of the items in the cart and displays it to the user. Input form included in 'checkout.ejs' where user can enter payment details and click purchase.
5	Use an object or an array in JavaScript;	'quan_price' object used in 'prodData.ejs', line 94 (addToCart function) to parse product quantity and price to JS object so that they can be stored in browser localStorage.
6	Use at least one custom module in Node	Auth1.js Node module created which includes function to add new users to website, and authenticate user credentials on login page.

7	Include capability for handling 'post' requests	'Post' request used in line 44 of 'app1.js' to handle user login and root user to '/login' if credentials are validated.
	Include capability for handling 'get' requests	'Get' request handled in lines 63 and 110 of 'app1.js'. One function used to root user to '/shop' and generate the product data page using the data in the MySQL database. Req number taken from P/N search on 'Home.html'. Req number also assigned to each of the product buttons on 'Home.html'. Another 'Get' method used to root user to '/cart' when 'checkout' button is pressed on any pages that include the button.
8	Include static content	'Home.html' is the static page served when the web server is started. 'login.html' and 'newsletter.html' also created for login page and newsletter sign up.
	Include dynamic content	'prodData.ejs' used to generate a page that injects database information into product data page depending on which product the user has selected to view.
9	Include the use of Templates in Node	'prodData.ejs' used as template for product data page. Product info injected into page through 'app1.ejs' depending on what item the user selects on 'Home.html'.
10	Include error messages to provide feedback to users in case of any issues or errors	Error messages used in 'app1.js' to print to console if there was an error connecting to the database, or if there was an error retrieving product data from database. Error messages also sent to console if user login failed. Error messages sent to user if invalid P/N is searched, and failed login page generated if login is unsuccessful
11	Connect to a MySQL database that contains relevant site information (eg., product info) using Node.	Database containing product info created. Connection to database created in 'app1.js', starting at line 17.
12	Use the bootstrap version 5 framework (via CDN)	Bootstrap styling in home page for image carousel, and page styling. Bootstrap styling also used across all other pages except for 'loginFailed.ejs'.